



Hateful tweet detection using a BiLSTM-BiGRU: An ensemble perspective

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ABSTRACT

Social media hate speech is an emerging issue, and there is a need to create automatic systems to identify and mitigate its effects. The rapid expansion of social media platforms, especially Twitter, has facilitated the dissemination of hate speech, presenting a major challenge for online communities. Such speech can have severe social and psychological consequences, including inciting violence, promoting extremism, and affecting mental health. Thus, it is essential to manage hateful content on Twitter. This paper presents an ensemble deep learning model that combines BiLSTM and BiGRU to enhance prediction accuracy and robustness. The model achieved 98.56% accuracy rate and demonstrated better generalization than existing methods, proving its effectiveness in identifying hate speech with fewer false positives. This paper offers a powerful tool for detecting and preventing harmful online behavior, contributing to a safer and more inclusive digital space.

1. Introduction

Social media is a prominent medium for the spread of hate speech as a result of the simplicity of anonymity, the ease of spreading quickly, and the effects of social and political tensions. It is possible to hide behind false accounts, encouraging them to upload hateful content with no risk of consequences in real life. Hate speech refers to any expression that promotes hatred, prejudice, or incites violence against an individual or group based on attributes such as sexuality, religion, race, gender, disability, or nationality. Increased use of online communication through media like the social media microblogging Twitter has increased how hate speech travels, and its becoming a contentious social issue is an indication that needs attention. With the global availability of the Internet for the last ten years, over 4.5 billion individuals, representing about 59% of the world's population [1], are now online. Social media is a significant draw, with sites such as Facebook, Twitter, and WeChat especially favored by the younger age group of users.

NLP [2] is a subfield of Artificial Intelligence and linguistics with the aim of making computers understand and interpret human language written text. Sentiment Analysis in NLP classifies sentiment in text, e.g., a review, comment, or social media update. Detection of hate speech is one of the main uses of NLP. This work intends to identify hate speech systematically produced on the Twitter Platform. Online forums are often filled with hostile or hateful content. To combat this, we are constructing a model to detect hate speech as follows:

- Twitter hate speech dataset is scraped from web, it is analyzed and different preprocessing steps are applied. This dataset is English-based and it contains hate speech on different concepts like race, religion, caste, communities, gender etc.
- Class imbalance is handled using Random Oversampling technique and most redundant and significant words are highlighted through word cloud.
- A novel ensemble approach is utilized in this work, which is a combination of BiLSTM and BiGRU, and attained an accuracy of 98.56%.

2. Literature survey

Hate speech detection has become a major problem due to the widespread of offensive content on social media. Initial studies primarily employed traditional ML approaches such as SVM [3], Random Forest, and Naïve Bayes [4], which depended on manually constructed features like TF-IDF, n-grams, and sentiment analysis. Nevertheless, these approaches tend to struggle with context and various forms of hate speech expression.

Roy et al. [1] introduced an LSTM-based DL model for Hate speech detection in real-time Twitter streams. The research utilizes TF-IDF for text vectorization and compares the performance of LSTM with different machine learning models, namely SVM, ANN, XGB, BERT, KNN, RF, NB, and LR. The data is extracted from Twitter tweets in real

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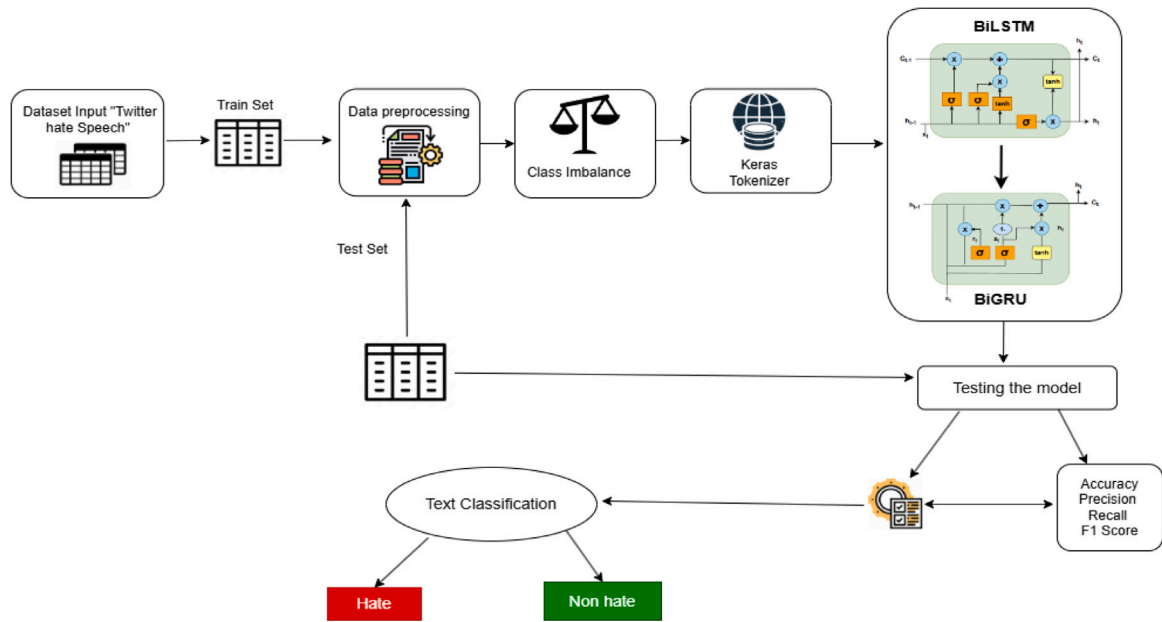


Fig. 1. Workflow of proposed methodology.

time using the Twitter API, and tweets are labeled as hate and non-hate speech. Experimental outcomes indicate that their proposed model performed well with an accuracy of 97%, surpassing other models.

Riyadi et al. [5] proposed a double-layer hybrid CNN-RNN model to enhance detection accuracy, especially when dealing with imbalanced datasets. The model combines CNNs and RNNs to utilize both local feature extraction and long-term dependency in text data. Oversampling methods are also used to alleviate data imbalance, which improves the performance of the model. The proposed approach exhibits better accuracy than traditional CNN- RNN methods, with 0.827 accuracy on imbalanced data and 0.908 accuracy on balanced data.

Koufakou et al. [6] designed an approach to tackle the problem of emotion detection benchmarking by merging several emotion-annotated text corpora into a single framework. The paper surveys 11 English-language datasets across diverse domains and projects them onto a common set of emotions drawn from Plutchik's and Shaver's models. The authors fine-tune a pre-trained transformer model (Distil Roberta) on individual datasets and their merged corpus and assess the effect on classification performance. Results indicate that training on the combined dataset enhances performance across a number of corpora, especially for minority emotions such as fear and surprise. The work emphasizes the difficulty of emotion annotation, variability issues in datasets, and the requirement for better generalization in NLP emotion detection.

Aslam et al. [7] presented a work, Emotion Detection and Sentiment Analysis on Cryptocurrency Related Tweets Using hybrid LSTM-GRU Model, which proposes a DL solution for emotion detection and sentiment identification of cryptocurrency-related tweets. The research utilizes a hybrid approach consisting of LSTM and GRU networks, with GRU being trained on LSTM-extracted features. Emotions are divided into happiness, sadness, anger, surprise, and fear through Text Blob and Text2Emotion. Different feature extraction techniques (TF-IDF, Word2Vec, and Bag of Words) and machine learning algorithms (SVM, Logistic Regression, etc.) are compared, with the developed LSTM-GRU ensemble registering 99% in sentiment analysis and 92% in emotion classification. The research points out the effect of cryptocurrency sentiment on market patterns and shows that deep learning ensembles perform better than conventional methods.

Ibrahim et al. [8] developed a work on Adaptive NN-Based Approach for Detecting Cyberbullying and Hate Speech in Social Media with Uncertainty Handling. This work introduces a new approach

to the detection of cyber bullying-related hate speech using Multi-Layer Perceptron (MLP) classification in conjunction with Neutrosophic Logic. The research deals with the problem of ambiguity and overlapping categories of cyberbullying detection by developing a fine-grained classification model that uses a one-against-one approach to multi-class classification. Results from experiments on cyberbullying datasets show the effectiveness of the model with 95%–97% accuracy, surpassing conventional machine learning methods like SVM and Random Forest. The research points out the necessity of managing class imbalance, adding data augmentation, and using deep learning methods like BERT in order to gain increased performance. The new framework offers a better and more adaptable method for cyberbullying detection in social media forensics.

Alrowias et al. [9] conducted a study on an Enhanced Cyberbullying Detection mechanism using a RoBERTa transformer model With GloVe Features presents a hybrid model based on the robustly optimized transformer model, RoBERTa, with GloVe word embeddings to detect cyberbullying on social media. The authors mention increasing incidents of cyberbullying on Twitter and social media and hence the requirement of an automated system to detect the same. They are suggesting a model that combines RoBERTa with different machine learning and deep learning methods such as Random Forest, SVM, BiLSTM, and CNN. The experimental evidence shows that their model based on RoBERTa performs better compared to conventional models with performance metrics: accuracy –95%, F1-score-96%, further is validated using k-fold cross-validation. This work focuses on the efficacy of transformer models in identifying cyberbullying and proposes enhancements to address class imbalance and multi-domain data in future work.

3. Proposed work

The proposed work is used for identifying hate speech in online communities, mainly on Twitter. A dataset is taken, which consists of tweets, and various preprocessing techniques are applied on the dataset. The data is balanced using random oversampling, i.e., by upsampling the minority class. Tweets are tokenized and padded, and sequences are returned. A sequential model is constructed and tested with a few tweets. Prediction metrics such as accuracy, precision, recall, and F1 score are used in this work. Each phase of the proposed work is represented in Fig. 1.

[illegible]

Fig. 2. Dataset image.

3.1. Dataset collection

The dataset is extracted from the Kaggle repository, it consists of 31 962 records, of which 29 720 tweets are hateful and 2242 tweets are non-hateful, it is mentioned in the below Fig. 2. The dataset is divided into a training set and a validation set. It is divided into 85% for training and 15% for validation. It contains a tweet column with the text and a label column for binary classification. Testing is done on unseen random tweets, simulating real-world application.

3.2. Data preprocessing

Data preprocessing is a vital step in NLP tasks. The tweet dataset used in this study contains various noisy elements, including URLs, special characters like hashtags and apostrophes, and irrelevant symbols. To ensure accurate analysis, it's essential to clean and preprocess the data by removing or normalizing these unwanted elements. In this step, the data is cleaned by

- Removing URLs
- Punctuations and Special Characters
- Converting text to lowercase to ensure uniformity.

After the preprocessing step, common words are identified from both the classes(class 0 and class 1). Words of class 1 and class 0 are represented using wordcloud technology in the below [Fig. 3](#).

3.3. Data sampling

There is a great imbalance in the dataset considered for this work. Class imbalance is a prevalent issue in text classification problems like hate speech detection. The Class Imbalance of dataset is represented in the form of a count plot in the below Fig. 4. To solve this problem, up-sampling is a common method whereby the minority class samples are replicated randomly until they match the count of the majority class samples. By using Random oversampling, the model gets an even representation of both classes (i.e., equal distribution of both the classes, as shown in below Fig. 5), which minimizes bias and enhances classification performance. With the use of up sampling, the dataset has a more balanced distribution, and the model can learn patterns from hate speech and non-hate speech instances effectively.

3.4. Tokenization and padding

Tokenization is performed by Keras Tokenizer, which transforms text data into numerical sequences by mapping words to distinct integer indices. A vocabulary size is set to 20,000 words to ensure that the model focuses only on important and relevant words in the dataset. Tweets are converted into sequences of numerical tokens using a tokenizer, which was trained on the dataset to map words to integer values. Following tokenization, the sequences were standardized to a uniform length of 75 words using padding, ensuring that all inputs had the same dimensionality for efficient processing. Finally, the dataset

is split into train and validation sets, with 85% of the data utilized for training and 15% used for validation, maintaining consistency and reproducibility through a fixed random seed.

3.5. Model architecture

The Proposed model architecture is an ensemble work leveraging the combination of two recurrent neural network variants. A sequential model is used, which is a combination of various layers represented in Fig. 6, where BiLSTM and BiGRU are combined sequentially. This model is trained with 10 epochs. The output of BiLSTM is normalized by using a Normalization Layer and sent as input to BiGRU. The output from the BiGRU layer is first normalized using Layer Normalization and then fed into a final BiLSTM layer, which generates the final output.

In this work, a total of 12 layers are used:

Hate Speech Detection using BiLSTM-BiGRU-BiLSTM Architecture Algorithm:

Input: Preprocessed tweet text T

Step 1: Tokenize and convert T to embedding sequence X

Step 2: Pass X through first BiLSTM layer: $H_1 = \text{BiLSTM}_1(X)$

Step 3: Pass H_1 through BiGRU layer: $H_2 = \text{BiGRU}(H_1)$

Step 4: Pass H_2 through second BiLSTM layer:

$$H_3 = \text{BiLSTM}_2(H_2)$$

Step 5: Pass through dense layers with dropout

Step 6: Final sigmoid layer outputs $\hat{y} \in [0, 1]$

Step 7: $\hat{y}$, Output obtained.

Output: If $\hat{y} > 0.5$ indicates hate speech else non hate speech

- **Embedding Layer:** The tokenized and padded sequences are given as input to this layer. Embedding layer converts words to dense vectors (represented by an integer index) of 300 dimensions, where similar words are located closer in the vector space. Embedding vectors are calculated using the below Eq. (1)

$$E = W_e[x_1, x_2, \dots, x_T],$$

$$\text{where } W_e \in \mathbb{R}^{V \times d} \quad (1)$$

- V : Vocabulary size
- d : Embedding dimensions
- T : Sequence length

- **Spatial Dropout Layer:** SpatialDropout1D layer randomly drops entire embedding vectors at each time step with a probability of 0.3. This layer reduces overfitting by dropping entire word embeddings, 30% of the words are randomly set to zero in every training step.
- **BiLSTM Layer:** BiLSTM model analyzes the input sequence in forward direction and backward directions using the Eqs. (2) (3) (4) (5) (6) (7), which are extracted from [10]. There are 512 hidden LSTM units total, of which 256 units are used in the forward direction, and 256 units are used in the backward direction. This layer is preferable because it is effective at learning long-range dependencies, such as when the hateful intent of a sentence is implied by distant words.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (2)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (3)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (4)$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (5)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (6)$$

$$h_t = o_t * \tanh(C_t) \quad (7)$$

- **Layer Normalization:** This layer normalizes each word's vector to have zero mean and unit variance. Each vector is normalized



Fig. 3. Words in class 0 and class 1.

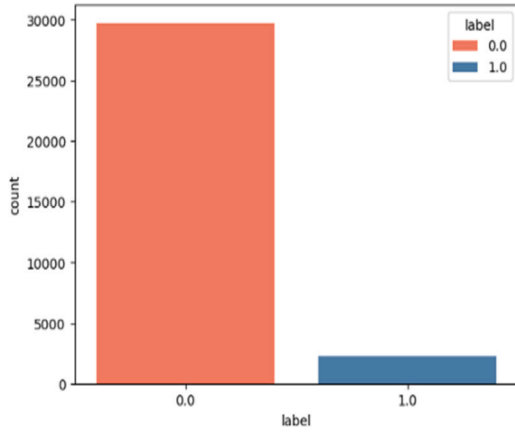


Fig. 4. The count of class 0 and class 1, before sampling the data.

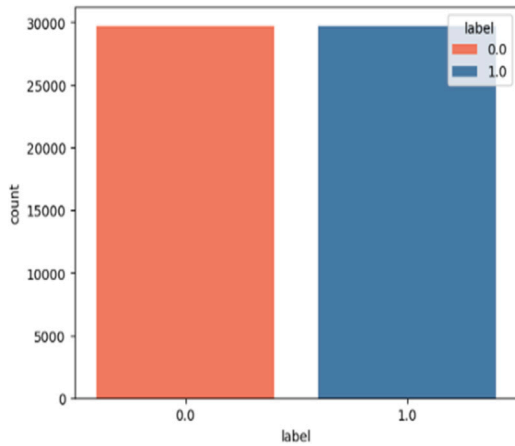


Fig. 5. The count of class 0 and class 1, after sampling the data.

in this phase by using the below formulae (8). It improves the effectiveness and performance of both LSTM and GRU outputs.

$$\hat{x} = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}} \cdot \gamma + \beta \quad (8)$$

- **BiGRU Layer:** BiGRU Layer captures additional dependencies faster than LSTMs; a total of 256 hidden GRU units are used. Each GRU direction contains 128 units, yielding an output shape of 256 per time step. GRUs use update and reset gates to control the flow of information, which helps in modeling shorter-term dependencies and complements the longer-term memory of the LSTM layer. It works based on the below formulas (9)–(12), these

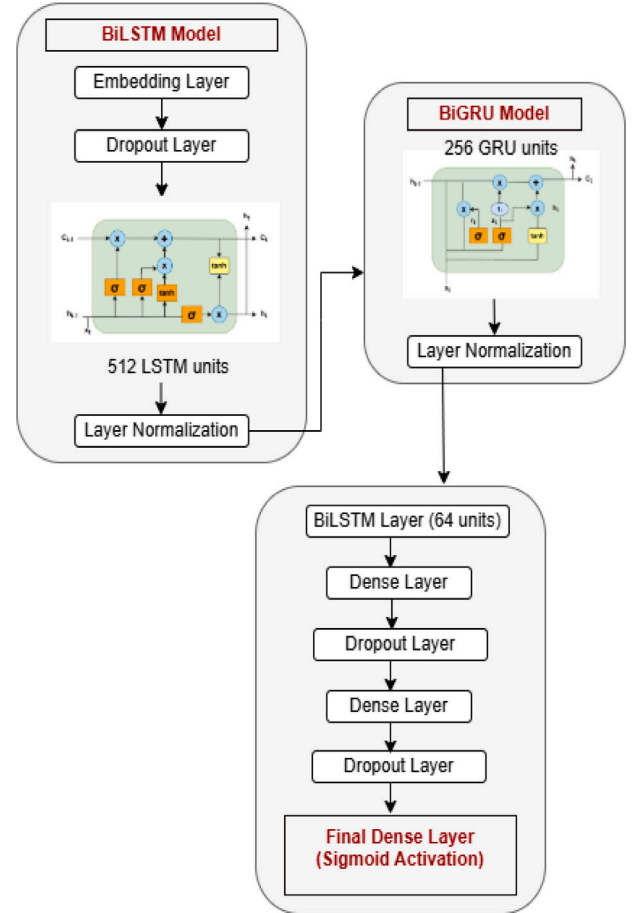


Fig. 6. Model architecture.

are extracted from [11].

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t]) \quad (9)$$

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t]) \quad (10)$$

$$\tilde{h}_t = \tanh(W_h \cdot [r_t * h_{t-1}, x_t]) \quad (11)$$

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t \quad (12)$$

- **Layer Normalization:** The outputs of the previous layer are normalized/ scaled in such a way that they attain a mean value of 0 and variance of 1 for every vector. It helps stabilize learning by avoiding spikes or extreme values and keeps each layer's input in a similar range.

- Another BiLSTM Layer: In order to get the final output, one more BiLSTM layer is used; from this layer, no sequences are returned. 64 hidden LSTM units are used for obtaining the final vector.
- Dense Layer: This layer applies a ReLU activation function on the final vector. The equation is given as (13). The 128 neurons help in transforming the sequential output into a form suitable for binary classification.

$$y = \text{ReLU}(W \cdot x + b) \\ = \max(0, W \cdot x + b) \quad (13)$$

- Dropout Layer: This layer randomly deactivates, 40% of the neurons during training to prevent overfitting by using the below formulae (14). It applies a random binary mask and zeroes out 40% of the neurons by using the formulae:

$$\tilde{x} = \frac{r \cdot x}{p} \quad (14)$$

- Dense Layer: This layer applies a ReLU Activation function on the obtained output. The 64 neurons help in transforming the sequential output into a form suitable for binary classification.
- Dropout Layer: This Dropout Layer again deactivates 30% of the neurons, adding another layer of regularization to reduce overfitting. It applies a random binary mask and zeroes out thirty percent of the neurons.
- Final Dense Layer: The Final layer is a fully connected layer with sigmoid activation function (Eq. (15)), outputs a single probability value between 0 and 1. This final output represents the likelihood that the input text contains hate speech. In our work, the threshold considered is 0.5, which is applied to determine the final binary classification.

$$\hat{y} = \sigma(W \cdot x + b) = \frac{1}{1 + e^{-(W \cdot x + b)}} \quad (15)$$

- Adam W Optimization: Adaptive Estimation with decoupled weight decay is used with a learning rate of 0.0003 to reduce overfitting by penalizing large weights (Eq. (16)). Binary Cross Entropy is a loss function used in this Classification problem, where the output obtained is either 0 or 1.
- Parameter update with decoupled weight decay:

$$\theta_t = \theta_{t-1} - \eta \left(\frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} + \lambda \cdot \theta_{t-1} \right) \quad (16)$$

4. Results

The evaluation metrics used for this work are Accuracy, Precision, Recall and F1 Score. The proposed work is compared against previous architectures used in different papers. The proposed architecture, BiLSTM with BiGRU, attained accuracy-98.56%, Precision-97.78%, Recall-99.42% and F1 score-98.59%. The efficiency of the proposed approach to detect hate speech was compared to other reported methods, including NB [12], LR [13], XGB [14], k-NN [15], ANN [16], LSTM [1] and LSTM+GRU [7]. Results are tabulated in Table 1, the proposed architecture obtained the highest accuracy. The evaluation metrics are calculated using the below confusion matrix Fig. 7:

The proposed model architecture is compared with various ML models and some ensemble works, the above mentioned are the comparison results 8. The curve closely follows the left and top edges of the plot, and the Area Under the Curve (AUC) is an impressive 0.9983 (Fig. 9), suggesting that the model achieves near-perfect discrimination capability. This high AUC value confirms the performance of the proposed hybrid DL model in detecting hate speech accurately.

Table 1

Performance comparison of various models.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 score (%)
Naïve Bayes	53.56	53.98	53.56	52.85
Logistic Regression	62.42	63.11	62.42	61.76
XG Boost	92.38	92.52	92.38	92.38
KNN	93.98	94.50	93.98	93.95
ANN	78.68	78.78	78.68	78.65
LSTM	59.11	59.73	59.18	59.45
LSTM+GRU	96.25	94.00	98.92	96.40
BiLSTM+BiGRU	98.56	97.78	99.42	98.59

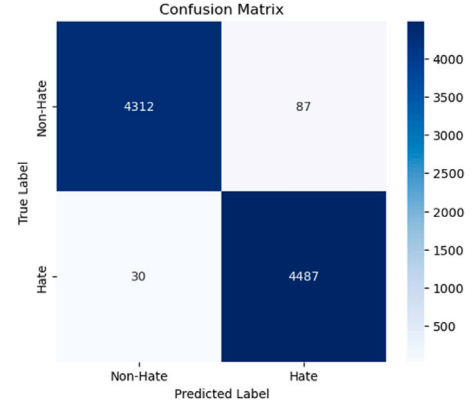


Fig. 7. Confusion matrix.

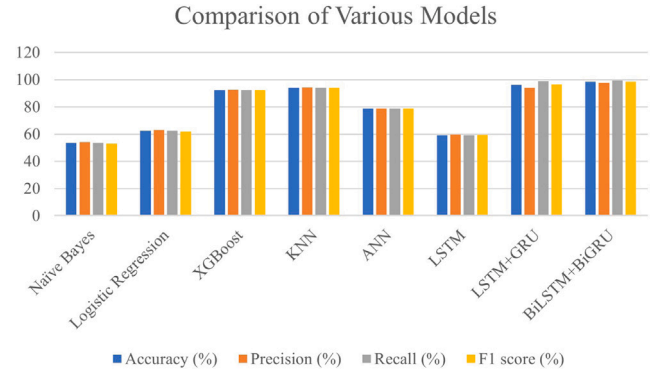


Fig. 8. Performance comparison of various ML and DL models.

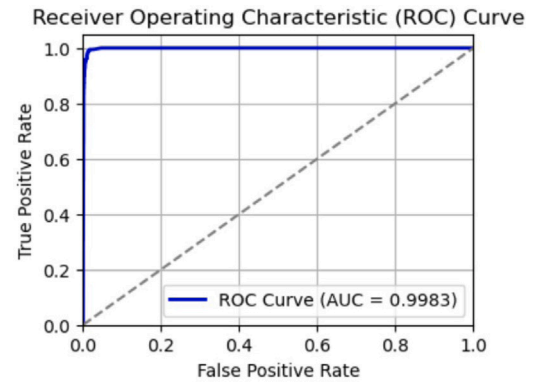


Fig. 9. Receiver Operating Characteristic (ROC) curve for the proposed model.

5. Conclusion

This paper presents an effective approach to detecting hate speech on Twitter using a Hybrid DL model. By combining Bidirectional LSTM and GRU layers, the proposed model successfully captures both forward and backward contextual information from the tweets. Pre-processing techniques such as removing URLs, punctuation, stop words, and applying lemmatization, combined with tokenization and embedding methods, helped in enhancing the quality of input data. The model was evaluated using prediction metrics such as accuracy, precision, recall, and F1-score, achieving a high validation accuracy of 98.56%, with a precision of 97.78%, recall of 99.42%, and F1-score of 98.59%. The results demonstrate the model's capability to distinguish between hateful content and non-hateful content effectively. The Model is not trained on multilingual data. It cannot handle sarcasm, words with various slang, etc. Future improvements may involve working on other Online Communities/Platforms. Future enhancement to this work could be to apply the proposed ensemble work on Multilingual datasets. The study contributes to ensuring safer digital communication platforms.

CRedit authorship contribution statement

Imandi Tejaswini: Supervision & Writing – review & editing & Writing – original draft. **Venkata Gayathri Ganivada:** Software, Resources, Methodology, Data curation, Conceptualization. **Appala Srinivasu Muttipati:** Writing – review & editing & Writing – original draft.

Declaration of competing interest

No conflict of interest

Data availability

Data will be made available on request.

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